

Overview of Statistical Learning and Modeling – 36-600

Project 2: Linear Regression
Due Friday, October 24, 11:59pm

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The goal of this project is to use linear regression to model the temperatures of stars given their brightness, their relative motion across the sky, and their location in the sky.

Your **HTML** file (generated by knitting an **R Markdown** source file) should be uploaded to Canvas by Friday, October 24th, at 11:59 PM. As was the case for Project 1, there is no limit on the length here, but concision is a virtue. (We will have class as usual on Thursday the 23rd, but there will be no accompanying lab due; this will help provide time to complete the project.)

Data

You will examine the dataset `stellar_temperature.csv`, which is downloadable from the Project 2 Canvas module page.

The response variable is `teff`, the effective temperature of a star in degrees Kelvin. Your goal is inference, but what that really means is that you will, at least, have to check for multicollinearity, compute variance-inflation factors, do best-subset selection, and look at PCA and PC regression. You will want to output a test-set mean-squared error calculated for a full linear model, the best-subset-selection linear model, and any others you find compelling. More “expectations” are listed out below.

The predictor variables are the following:

name	description
<code>ra_x</code>	right ascension (celestial longitude)
<code>dec_x</code>	declination (celestial latitude)
<code>parallax</code>	stellar parallax (inversely proportional to distance)
<code>pmra</code>	proper motion along longitude line
<code>pmdec</code>	proper motion along latitude line
<code>[g, b, r]_mag</code>	magnitudes in green, blue, and red bands
<code>br_col</code>	the B-R color
<code>L</code>	galactic longitude
<code>B</code>	galactic latitude

Expectations

Your report should include the following elements:

- A description of the data (sample size, number of variables), and a list of removed variables, if there are any.

- Concise EDA, including the identification and removal of proposed outliers (if there are any) and, potentially, the transformation of the response variable (but first see below). (*Note*: always try transformations first before identifying and removing outliers – that lonely data point in a highly skewed distribution may look completely fine after a transformation is performed.) You can, if you wish, transform any predictor variables that are highly skewed, but this is not required. Also, create a correlation plot and comment on the possibility of multicollinearity in the predictor variables.
- A description of how you split the data into training and test sets.
- An analysis of the full dataset with linear regression, with comments on the output (does the fit appear to be good?). Provide the mean-squared error (and the root mean-squared error, with an interpretation of this number!) and the predicted response vs. observed response diagnostic plot (being sure to make the limits the same along each axis, and being sure to overlay a diagonal line with slope 1).
- Brief comments on the value of the adjusted R^2 : is the linear model useful, or might other models perform better?
- A potential consideration of response transformations; realize that the response variable needs to be transformed if and only if (a) the *residuals* of the fit are not normally distributed *and* (b) you wish to do inference using the hypothesis test output from `lm()`. To plot the residuals, find the difference between the observed test-set response values and the predicted test-set response values, and make a histogram: does this plot look Gaussian? If you perform a hypothesis test, do you reject the null hypothesis that the residual distribution is plausibly normal?
- Does a plot of the residuals versus the predicted test-set response values suggest that the variance around the regression line is σ^2 ?
- Computation of variance-inflation factors. Comment if it appears that there is substantial multicollinearity, or not. Identify the potentially problematic variables (just list them). Do not automatically start removing variables with high vif factors, if there are any... first study the results of variable selection procedures, e.g., best-subset selection.
- A best-subset-selection analysis of the dataset. Which predictor variables are important for predicting stellar temperatures? Compute the MSE for the `BestModel` and compare it to the MSE for the full set of predictors. (And remember: you need to change the name of the response variable to `y` before running BSS! See the snippet below.) Do you expect to see a large number of variables removed? A small number? Do the results match your expectation?

```
w <- which(names(df) == "teff")
y <- df[, w]
df <- df[, -w]
df <- data.frame(df, "y" = y)
```

- Do a principal components analysis on the predictors with the goal of determining the “true dimensionality” of the predictor variables. Select some number of PCs to keep (while being sure to indicate what criterion you used). Run linear regression on these PCs and compute the test-set MSE, and show how it compares with the full-set MSE and the BSS MSE. (The expectation, given the example in the PCA notes, is that the MSE will increase, perhaps substantially, if we do not include all the PCs in our PC regression model.) It’s not *necessary* to comment on how the original predictors map to PC space, especially if that mapping is particularly convoluted – you might want just to pick a suitable number of PCs and proceed directly to regression (though, you might not).
- *Note*: as there are no factor variables in the dataset, you won’t need to pursue any random effects analyses in this project.

```
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2     4.0.0      v tibble     3.2.1
## v lubridate  1.9.3      v tidyr      1.3.1
## v purrr       1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
```

```
df<- read.csv("C:/Users/Trish/Documents/CMU/CMU-3660/Projects/stellar_temperature.csv")
y <-df$teff
summary(df)
```

```
##      teff          ra_x          dec_x          parallax
## Min.   :3664   Min.    : 0.0043   Min.    :-85.713   Min.    : 0.1179
## 1st Qu.:4911   1st Qu.: 80.6802   1st Qu.: 9.583   1st Qu.: 0.6239
## Median :5548   Median :131.7663   Median : 25.928   Median : 1.0460
## Mean   :5518   Mean   :154.5872   Mean    : 23.149   Mean    : 1.4354
## 3rd Qu.:6018   3rd Qu.:231.4918   3rd Qu.: 39.772   3rd Qu.: 1.7423
## Max.    :8631   Max.    :359.9147   Max.     : 85.299   Max.     :16.9419
##      pmra          pmdec          g_mag          b_mag
## Min.   :-258.5010   Min.    :-207.1520   Min.     : 5.191   Min.     : 5.776
## 1st Qu.: -5.9387   1st Qu.: -9.3328   1st Qu.:12.777   1st Qu.:13.207
## Median : -0.9095   Median : -4.0930   Median :13.878   Median :14.322
## Mean    : -1.6585   Mean    : -6.0106   Mean     :13.831   Mean     :14.302
## 3rd Qu.: 3.1290   3rd Qu.: -0.5092   3rd Qu.:14.983   3rd Qu.:15.481
## Max.     :188.1720   Max.     :106.0590   Max.     :18.809   Max.     :19.448
##      r_mag          br_col          L          B
## Min.    : 4.564   Min.    :-0.08625   Min.     : 0.0385   Min.    :-89.23
## 1st Qu.:12.204   1st Qu.: 0.33404   1st Qu.: 91.5764   1st Qu.: -16.91
## Median :13.287   Median : 0.42862   Median :163.1523   Median : 12.25
## Mean     :13.222   Mean     : 0.47147   Mean     :154.3933   Mean     : 11.80
## 3rd Qu.:14.362   3rd Qu.: 0.57561   3rd Qu.:198.4155   3rd Qu.: 37.98
## Max.     :18.689   Max.     : 2.27454   Max.     :359.9994   Max.     : 89.42
```

```
dim(df)
```

```
## [1] 10000    12
```

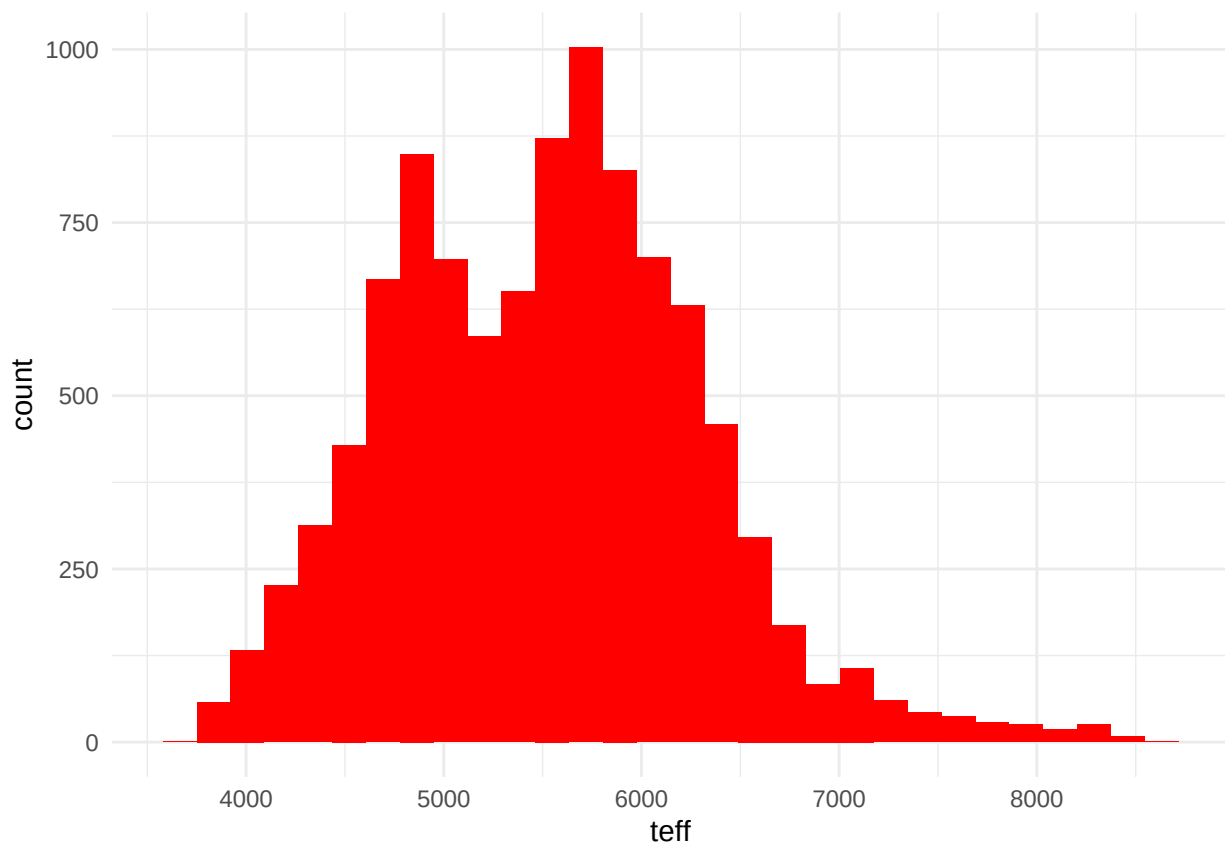
```
str(df)
```

```
## 'data.frame':    10000 obs. of  12 variables:
## $ teff      : num  6112 5173 6122 6266 6717 ...
```

```
## $ ra_x      : num  239.7 39.8 143.6 118.6 291.6 ...
## $ dec_x     : num  23.18 36.76 9.22 16.17 39.22 ...
## $ parallax: num  1.304 3.606 1.163 0.979 1.646 ...
## $ pmra      : num  -7.268 25.877 10.092 -0.166 5.901 ...
## $ pmdec     : num  2.029 -172.907 -10.563 -9.018 0.276 ...
## $ g_mag     : num  13.7 13.1 13.8 12.9 11.3 ...
## $ b_mag     : num  14 13.6 14.1 13.1 11.6 ...
## $ r_mag     : num  13.2 12.5 13.4 12.5 11 ...
## $ br_col    : num  0.319 0.463 0.277 0.267 0.219 ...
## $ L         : num  38.4 145.8 224.4 205 71.8 ...
## $ B         : num  47.6 -21.3 40.3 21 10.6 ...
```

The data set has 10000 observations of 12 variables.

```
ggplot(df, aes(teff)) + geom_histogram(bins = 30, fill = "red") + theme_minimal()
```



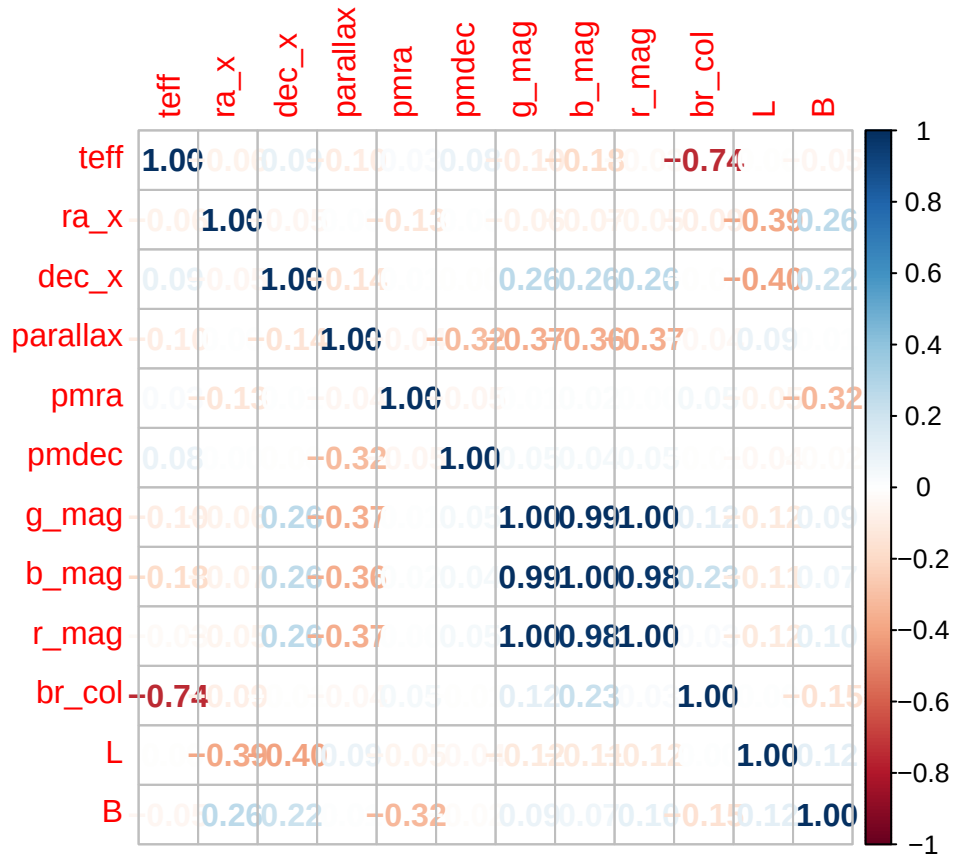
The data appears to be bimodal, 1 mode possibly for colder star, 1 mode for hotter stars. This indicates to me that the data could be grouped into at least 2 groups. The data is also slightly skewed, but since there is a large number of observations, we don't have to worry too much about the skew so no need to really transform/standardize the data.

```
library(corrplot)
```

```
## Warning: package 'corrplot' was built under R version 4.4.3
```

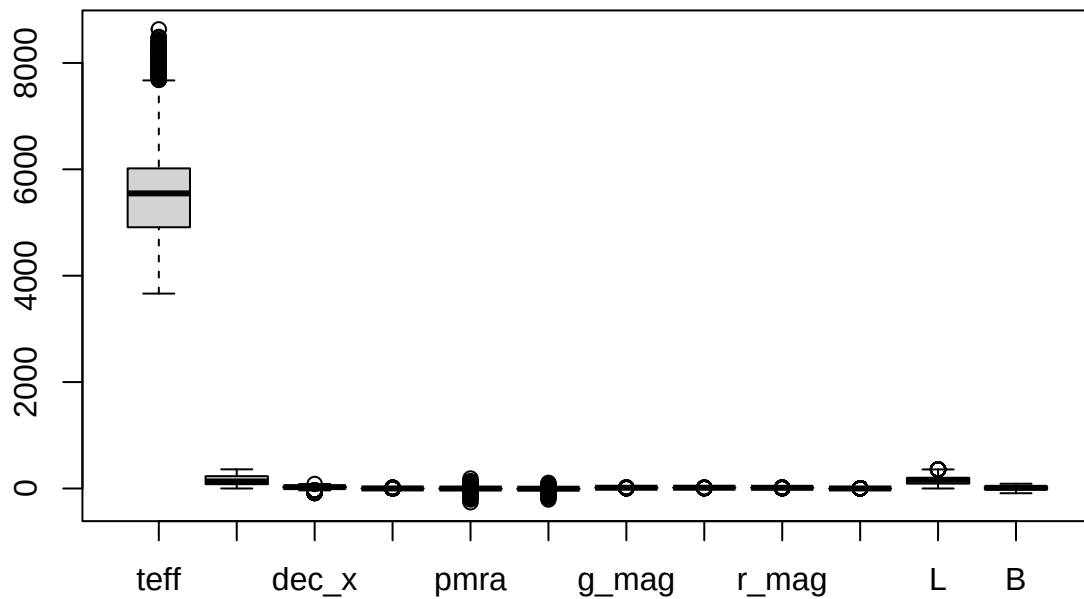
```
## corrplot 0.95 loaded
```

```
corrplot(cor(df), method = "number", tl.cex = 1)
```



From the correlation matrix, we can see that the mags are highly, positively, correlated to each other, while teff and br_col and teff are negatively correlated.

```
#Outliers
boxplot(df)
```



There are no outliers in the dataset except in teff.

```
#linear regression model
```

```
set.seed(42)
```

```
n <- nrow(df)
```

```
train_idx <- sample(1:n, 0.8 * n)
```

```
train_set <- df[train_idx, ]
```

```
test_set <- df[-train_idx, ]
```

```
lm_full <- lm(teff ~ ., data = train_set)
```

```
summary(lm_full)
```

```
##
```

```
## Call:
```

```
## lm(formula = teff ~ ., data = train_set)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
```

```
## -1137.3 -276.0   -87.2   147.8  4360.3
```

```
##
```

```
## Coefficients: (1 not defined because of singularities)
```

```
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)  8.278e+03  6.320e+01 130.973  < 2e-16 ***
```

```
## ra_x         -4.186e-01  6.461e-02  -6.479 9.80e-11 ***
```

```
## dec_x         5.011e+00  2.738e-01 18.298  < 2e-16 ***
```

```
## parallax     -6.903e+01  4.269e+00 -16.170  < 2e-16 ***
```

```
## pmra         -2.045e-01  3.641e-01  -0.562  0.574
```

```
## pmdec      1.860e+00  4.166e-01  4.464 8.17e-06 ***
## g_mag     -6.512e+03  3.947e+02 -16.499 < 2e-16 ***
## b_mag      1.107e+03  1.790e+02  6.182 6.63e-10 ***
## r_mag      5.407e+03  2.184e+02 24.759 < 2e-16 ***
## br_col      NA      NA      NA      NA
## L          6.286e-01  9.006e-02  6.980 3.20e-12 ***
## B          -4.272e+00  1.757e-01 -24.322 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 450.8 on 7989 degrees of freedom
## Multiple R-squared:  0.6611, Adjusted R-squared:  0.6607
## F-statistic: 1559 on 10 and 7989 DF, p-value: < 2.2e-16
```

To split the dataset, I did a 80 20 split for training and testing. Based on the linear regression model produced above, I suspect the model is not that great to determine how hot a star is because there is colinearity since `br_col` is from subtracting the `b_mag` and `r_mag` columns. The adjusted r^2 value is around 66% so the model explains about 66% of the data, but i think it can be greatly improved.

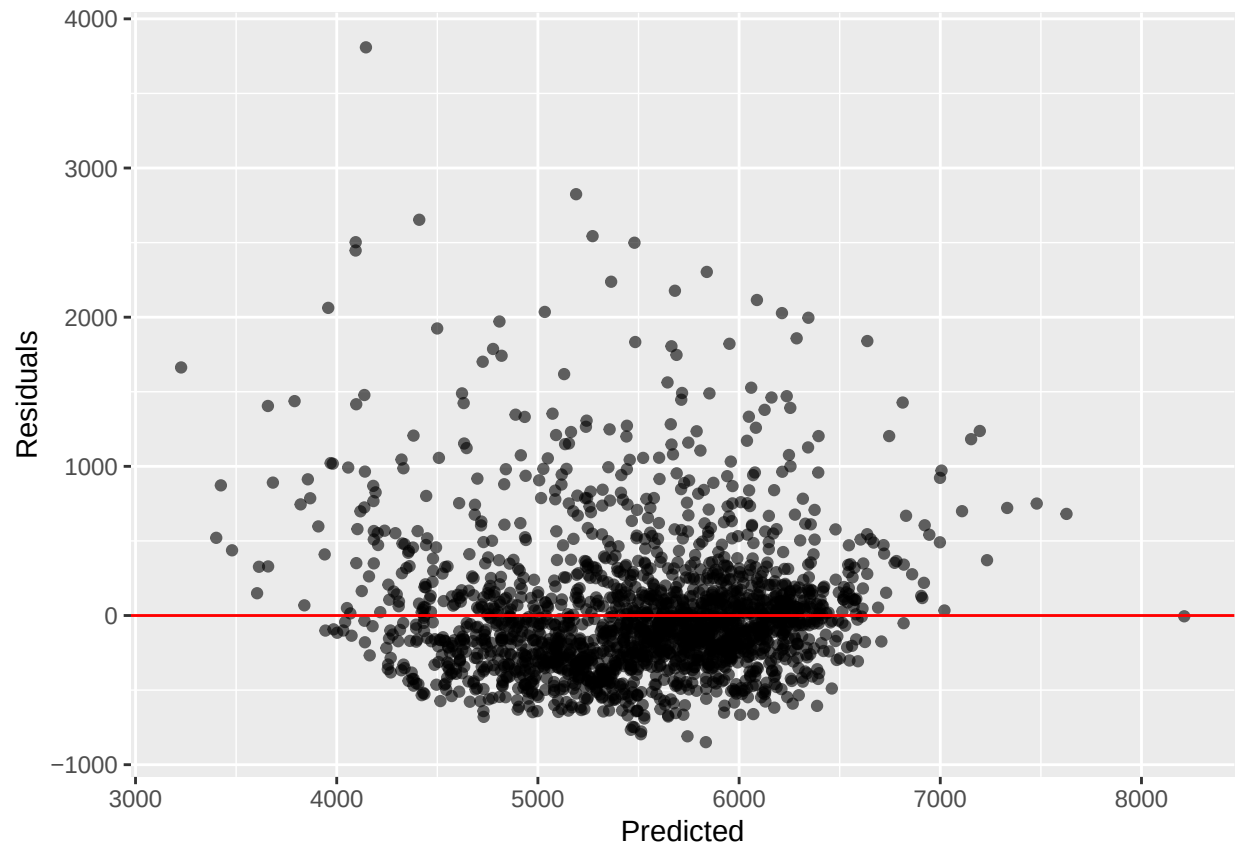
```
# MSE & RMSE
pred_full <- predict(lm_full, newdata = test_set)
mse_full <- mean((pred_full - test_set$teff)^2)
rmse_full <- sqrt(mse_full)
print(rmse_full)
```

```
## [1] 468.2549
```

```
print(mse_full)
```

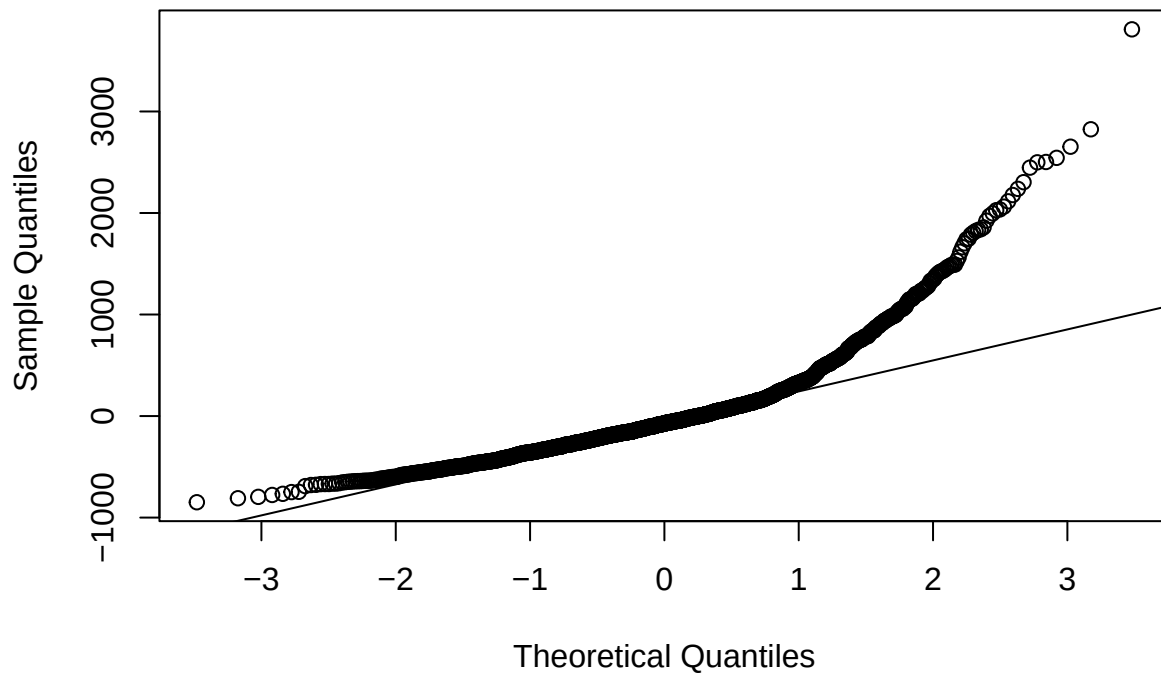
```
## [1] 219262.7
```

```
# Residuals plot
resid_full <- test_set$teff - pred_full
ggplot(data.frame(pred_full, resid_full),
  aes(pred_full, resid_full)) +
  geom_point(alpha = 0.6) +
  geom_hline(yintercept = 0, color = "red") +
  labs(x = "Predicted", y = "Residuals")
```



```
# Normality  
qqnorm(resid_full); qqline(resid_full)
```


Normal Q-Q Plot



```
shapiro.test(resid_full)
```

```
##  
##  Shapiro-Wilk normality test  
##  
## data:  resid_full  
## W = 0.84205, p-value < 2.2e-16
```

The RMSE from the model is 468.2549. The MSE from the model is 219262.7. Based on the Residuals plot, we see there is some heteroscedacity so the data is not equally varied throughout.

```
library(car)
```

```
## Warning: package 'car' was built under R version 4.4.3  
  
## Loading required package: carData  
  
## Warning: package 'carData' was built under R version 4.4.3  
  
##  
## Attaching package: 'car'  
  
## The following object is masked from 'package:dplyr':  
##  
##   recode
```

```
## The following object is masked from 'package:purrr':
##
##      some
```

```
#vif(lm_full)
lm_full <- lm(teff ~ ra_x + dec_x + parallax + pmra + pmdec +
              g_mag + b_mag + r_mag + L + B, data = train_set)
vif(lm_full)
```

```
##      ra_x      dec_x    parallax      pmra      pmdec      g_mag
##  1.551893  1.578235  1.319552  1.129672  1.135975 16706.958587
##      b_mag      r_mag          L          B
## 3563.143063 4989.153901  1.787806  1.529802
```

At first vif did not work because of the br_col variable (it is a colinear variable). So I redid the model without it. Based on the new cif, we still see hi collinearity between the mag columns.

```
# Prepare data
library(leaps)
```

```
## Warning: package 'leaps' was built under R version 4.4.3
```

```
w <- which(names(train_set) == "teff")
y <- train_set[, w]
train_mod <- data.frame(train_set[, -w], y = y)
bss <- regsubsets(y ~ ., data = train_mod, nvmax = ncol(train_mod) - 1)
```

```
## Warning in leaps.setup(x, y, wt = wt, nbest = nbest, nvmax = nvmax, force.in =
## force.in, : 1 linear dependencies found
```

```
## Reordering variables and trying again:
```

```
bss_summary <- summary(bss)

# Identify best model by adjusted R^2 or Cp
which.max(bss_summary$adjr2)
```

```
## [1] 9
```

```
best_vars <- names(coef(bss, which.max(bss_summary$adjr2)))[-1]
formula_best <- as.formula(paste("teff ~", paste(best_vars, collapse = "+")))
lm_best <- lm(formula_best, data = train_set)

pred_best <- predict(lm_best, newdata = test_set)
mse_best <- mean((pred_best - test_set$teff)^2)
rmse_best <- sqrt(mse_best)
print(mse_best)
```

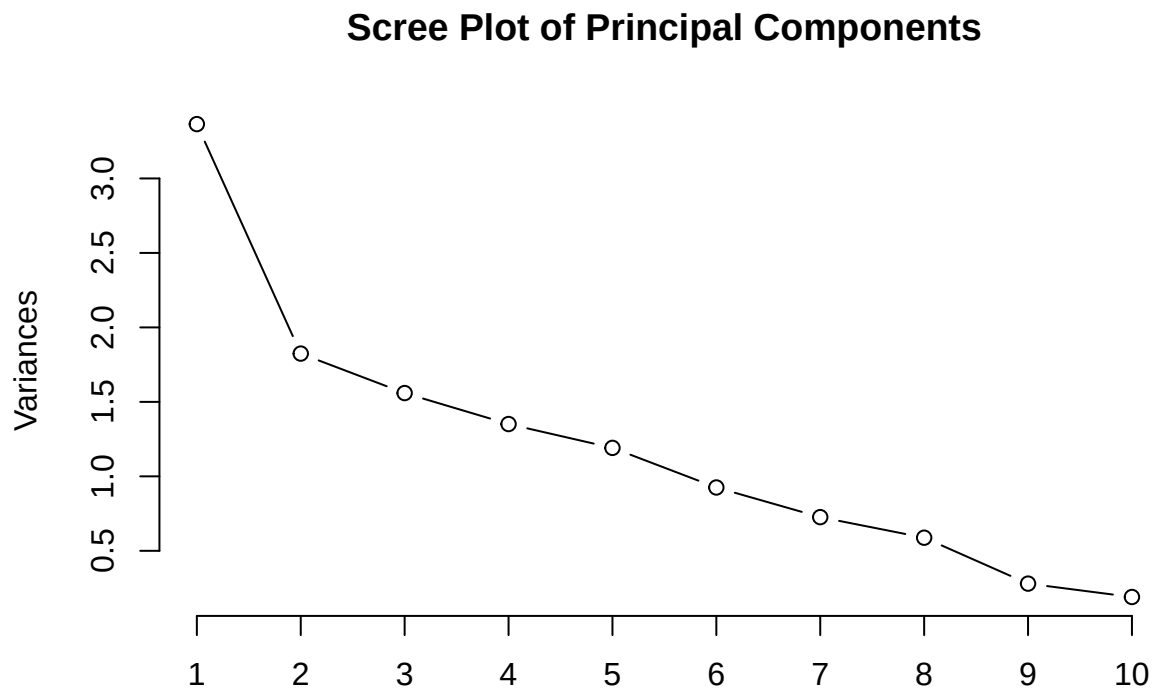
```
## [1] 219241.1
```

```
print(rmse_best)
```

```
## [1] 468.2319
```

Based on the mse, the BSS's mse is pretty close to the mse of the original model

```
pca_fit <- prcomp(train_set, center = TRUE, scale. = TRUE)
plot(pca_fit, type = "l", main = "Scree Plot of Principal Components")
```



Running PCA on the data, shows that there are 2 principal components in the data.

```
train_pc <- as.data.frame(pca_fit$x[, 1:2])
train_pc$teff <- train_set$teff

# For test set, project onto same PCs
test_scaled <- scale(test_set, center = pca_fit$center, scale = pca_fit$scale)
test_pc_scores <- as.data.frame(test_scaled %*% pca_fit$rotation[, 1:2])

# Fit regression model on first k PCs
pcr_mod <- lm(teff ~ ., data = train_pc)
summary(pcr_mod)
```

```
##
## Call:
## lm(formula = teff ~ ., data = train_pc)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1339.95  -265.94   -38.01   219.73  2818.27
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 5513.511      4.329  1273.6  <2e-16 ***
## PC1          57.825      2.360    24.5  <2e-16 ***
## PC2         489.923      3.206   152.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 387.2 on 7997 degrees of freedom
## Multiple R-squared:  0.7497, Adjusted R-squared:  0.7497
## F-statistic: 1.198e+04 on 2 and 7997 DF,  p-value: < 2.2e-16

# Predictions and MSE
test_preds_pcr <- predict(pcr_mod, newdata = test_pc_scores)
test_mse_pcr <- mean((test_set$teff - test_preds_pcr)^2)
test_rmse_pcr <- sqrt(test_mse_pcr)
print(test_mse_pcr)

## [1] 159427.6
```

The test set mse is much lower compared to both the BSSMSE and the original model's MSE.